

Your PhD Proposal Title

Your Name

June 11, 2026

Contents

1	Literature Review	2
1.1	Introduction	2
2	Continuum Model	3
2.1	Introduction	3
3	Molecular Dynamics	4
3.1	Introduction	4
4	Sensitivity Analysis	5
4.1	Introduction	5
4.2	Notation and Problem Setup	5
4.3	Local (Deterministic) Methods	5
4.3.1	One-at-a-Time (OAT) Methods and Tornado Diagrams	5
4.3.2	Differentiation-Based Methods	5
4.3.3	Screening Methods	6
4.4	Global (Probabilistic) Methods	6
4.4.1	Regression-Based Methods	6
4.4.2	Variance-Based Methods (Sobol Indices)	6
4.4.3	Density- and Moment-Independent Methods	6
4.4.4	Value of Information-Based Methods	6
4.5	Estimation and Computational Strategies	7
4.6	Sensitivity Settings and Interpretation	7
4.7	Choice of Method for This Work	7

Chapter 1

Literature Review

1.1 Introduction

Content here...

Chapter 2

Continuum Model

2.1 Introduction

Content here...

Chapter 3

Molecular Dynamics

3.1 Introduction

Content here...

Chapter 4

Sensitivity Analysis

4.1 Introduction

- What sensitivity analysis is and why it matters
- Role in model building, validation, and communication
- Scope: local vs. global, deterministic vs. probabilistic

4.2 Notation and Problem Setup

- Model as input-output mapping:
- Input space, output space
- Sensitivity measure vs. importance measure distinction

4.3 Local (Deterministic) Methods

4.3.1 One-at-a-Time (OAT) Methods and Tornado Diagrams

- Finite difference approach
- Individual effects, interaction effects, total effects
- Generalized Tornado diagrams

4.3.2 Differentiation-Based Methods

- Partial derivatives and comparative statics
- Differential importance measure (DIM)

- Elasticity

4.3.3 Screening Methods

- Sequential bifurcation
- Morris elementary effects method

4.4 Global (Probabilistic) Methods

4.4.1 Regression-Based Methods

- Standardized Regression Coefficients (SRC)
- Pearson and Spearman correlation
- Limitations: R^2 and nonlinearity

4.4.2 Variance-Based Methods (Sobol Indices)

- Functional ANOVA decomposition
- First-order index (main effect)
- Total-order index
- Higher-order interaction indices

4.4.3 Density- and Moment-Independent Methods

- Delta sensitivity measure (δ_i)
- Kullback–Leibler divergence-based measures
- Transformation invariance property

4.4.4 Value of Information-Based Methods

- Partial Expected Value of Perfect Information (EVPPI)
- Decision sensitivity vs. value sensitivity

4.5 Estimation and Computational Strategies

- Design-based: Sobol sequences, FAST, RBD
- Given-data (post-processing) approaches
- Meta-models / emulators (Gaussian Process, PCE)

4.6 Sensitivity Settings and Interpretation

- Factor prioritization setting
- Factor fixing setting
- Model structure setting
- Direction of change setting
- Stability/robustness setting

4.7 Choice of Method for This Work

- Justify which method(s) you will use and why
- Link to your specific model (continuum/MD)